```r
load("synthetic_raw.Rda")
library(SmoothHOOI)
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
```

This is an illustration of how to use our functions on synthetic data

```r
set.seed(1234)
```

Generate synthetic data

```r
synthetic <- sim_data2(L_tilde, R_tilde, mean_G, cov_G, E, p=207, noise_level=1, pattern = "structured"
synthetic_dat <- synthetic$sim_Mmiss

save(synthetic_dat, file = "synthetic_data.Rda")
```

Make the second order difference matrix

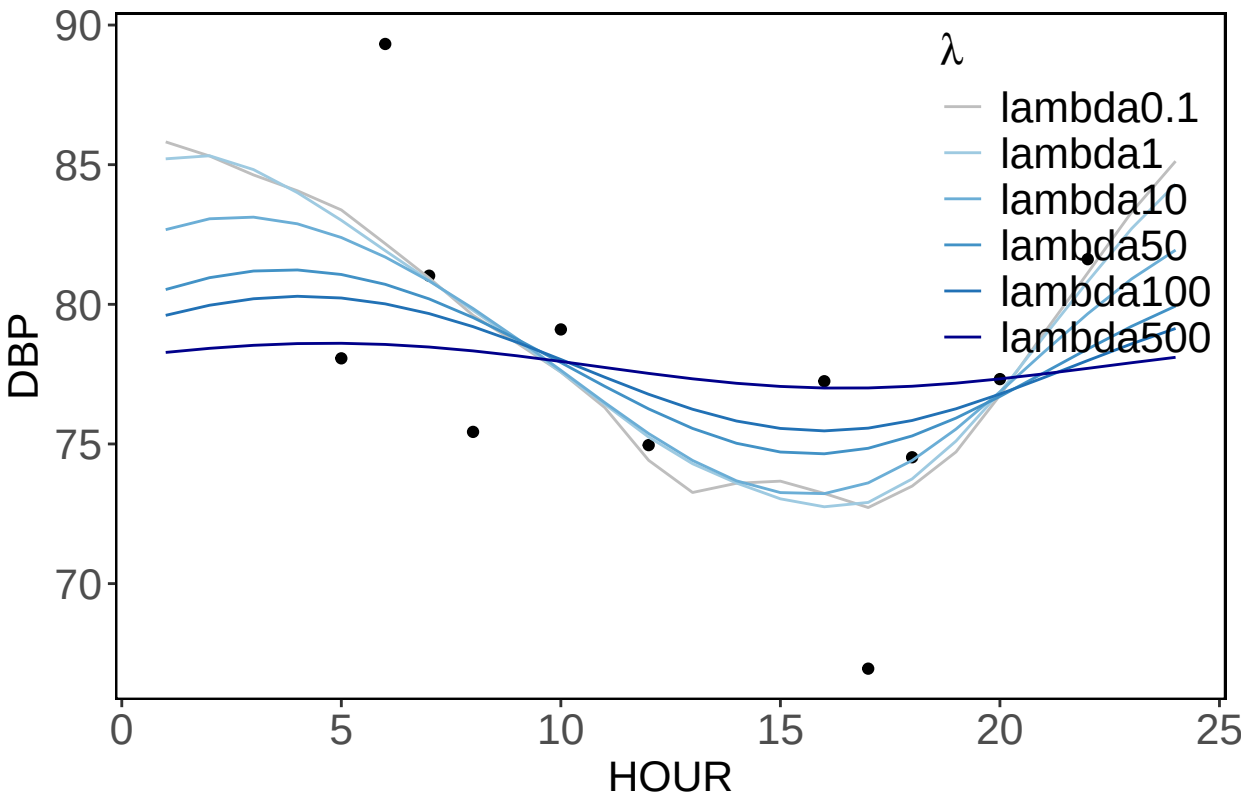
```r
D2 <- SecDiffMat(24)
```

Coarse-to-fine grid search (Example)

```r
original <- (synthetic_dat@data[ , 1, 180])*12+70
lambda_seq <- c(0.1, 1, 10, 50, 100, 500)
est_df <- matrix(NA, nrow=24, ncol=length(lambda_seq))
for (i in 1:length(lambda_seq)){
  lambda <- lambda_seq[i]
  res <-  mglram(tnsr = synthetic_dat@data, ranks = c(6, 3), init=0, D = D2,
                 lambda = lambda, max_iter = 500, tol = 1e-5, L0 = NULL)
  est_df[,i] <- (res$est[,1,180])*12 +70
```

```r
}
org_df <- data.frame(cbind(c(1:24), original))
est_df <- data.frame(cbind(c(1:24), est_df))
colnames(org_df) <- c("hour", "original")
colnames(est_df) <- c("hour", paste0("lambda",lambda_seq))
est_long <- pivot_longer(est_df, cols = starts_with("lambda"),
                         names_to = "lambda", values_to = "y")
est_long$lambda <- factor(est_long$lambda, levels=c("lambda0.1","lambda1","lambda10","lambda50","lambda
```

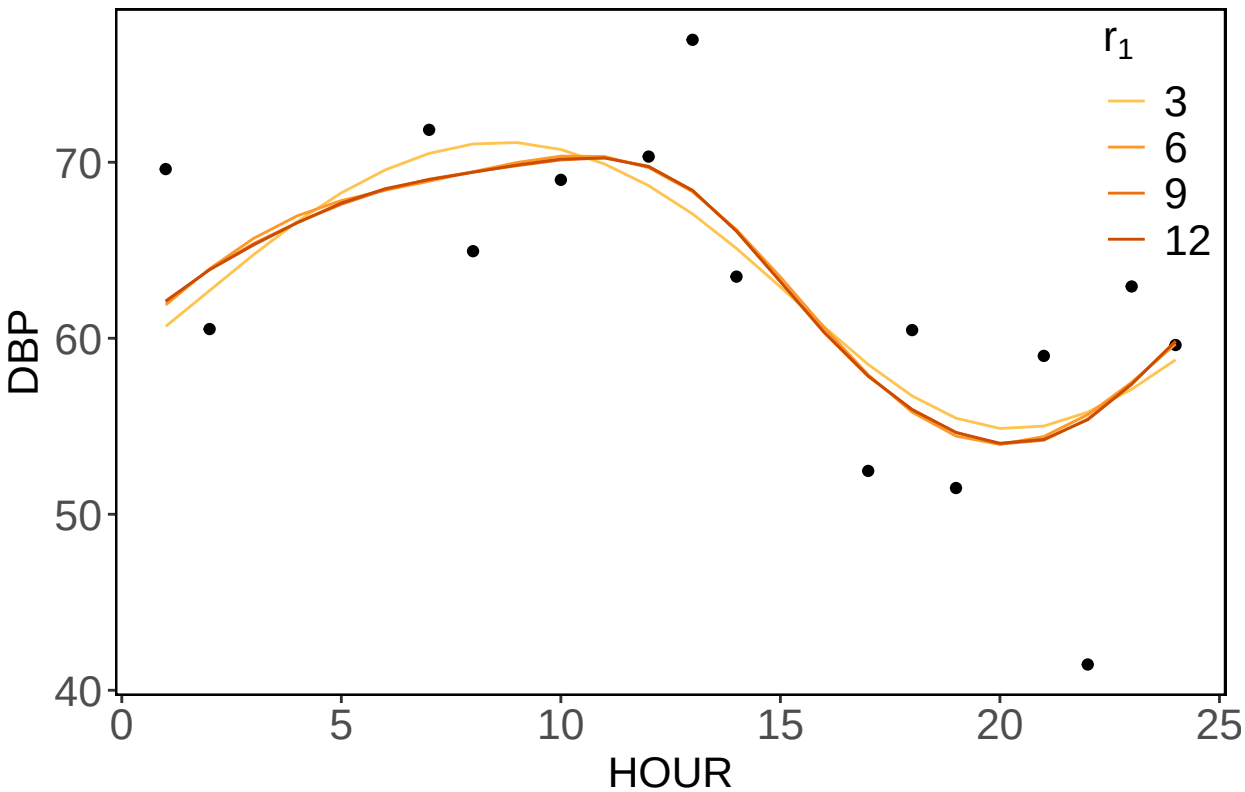
```r
custom_colors <- c("grey","#9ECAE1","#6BAED6", "#4292C6", "#2171B5","darkblue")
names(custom_colors) <- levels(factor(est_long$lambda))

ggplot2::ggplot() +
  geom_point(data = org_df, aes(x = hour, y = original), color = "black") +  # Original data
  geom_line(data = est_long, aes(x = hour, y =y, color = lambda)) +  # Estimated curves
  labs(title = "",
       x = "HOUR", y = "DBP") +
  scale_color_manual(values = custom_colors,
                     name = expression(lambda)) +
  theme(legend.position = c(1, 1),
    legend.justification = c(1, 1),
    panel.grid = element_blank(),
    panel.background = element_blank(),
    plot.background = element_blank(),
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.8),  # add border rectangle
    axis.line = element_blank(),
    legend.background = element_blank(),  # removes the outer background & border
    legend.key = element_blank(),
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 16),
    legend.text = element_text(size = 16,hjust=0),
    legend.title = element_text(size = 16)
    )
```




```
original <- (synthetic_dat@data[ , 1, 1])*12 + 70
r1_seq <- c(3,6,9,12)
est_df <- matrix(NA, nrow=24, ncol=length(r1_seq))

for (i in 1:length(r1_seq)){
  r1 <- r1_seq[i]
  res <-  mglram(tnsr = synthetic_dat@data, ranks = c(r1, 3), init=0, D = D2,
                 lambda = 10, max_iter = 500, tol = 1e-5, L0 = NULL)
  est_df[,i] <- (res$est[,1,1])*12 + 70
}

colnames(est_df) <- paste0("r", r1_seq)
est_df <- cbind(hour = 1:24, est_df)
est_df <- data.frame(est_df)
org_df <- data.frame(hour = 1:24, original = original)
est_long <- pivot_longer(est_df, cols = starts_with("r"),
                         names_to = "r", values_to = "y")
est_long$r <- factor(est_long$r, levels = paste0("r", r1_seq))
```

```
custom_colors <- c("#FEC44F", "#FE9929","#EC7014","#CC4C02")
names(custom_colors) <- levels(factor(est_long$r))

ggplot2::ggplot() +
  geom_point(data = org_df, aes(x = hour, y = original), color = "black") +  # Original data
  geom_line(data = est_long, aes(x = hour, y =y, color = r)) +  # Estimated curves
  labs(title = "",
       x = "HOUR", y = "DBP") +
  scale_color_manual(values = custom_colors,
```

```
                    labels = c("r3" = 3, "r6" = 6,
                               "r9" = 9, "r12" = 12),
                  name =  expression(r[1])) +
  theme(legend.position = c(1, 1),
    legend.justification = c(1, 1),
    panel.grid = element_blank(),
    panel.background = element_blank(),
    plot.background = element_blank(),
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.8),  # add border rectangle
    axis.line = element_blank(),
    legend.background = element_blank(),  # removes the outer background & border
    legend.key = element_blank(),
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 16),
    legend.text = element_text(size = 16,hjust=0),
    legend.title = element_text(size = 16)
    )
```



Find optimal hyperparameter using k-fold cross-validation

```
start_time <- Sys.time()

kcv_res <- kcv(tnsr=synthetic_dat@data, rank_grid=as.matrix(expand.grid(r1<-seq(3,6,by=1), r2<-c(2,3)))
               k=5, L0 =NULL, D=D2, tol=0.01, max_iter=500, init=0)
```

```r
end_time <- Sys.time()

exec_time <- end_time - start_time
print(exec_time)
```

```
## Time difference of 5.862901 mins
```

```r
kcv_res
```

```
## $MSE_mat
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 0.4797804 0.4757447 0.4733624 0.4716788 0.4703872 0.4693533 0.4684931
## [2,] 0.5189747 0.5055174 0.4978910 0.4927867 0.4890443 0.4861874 0.4839173
## [3,] 0.5293766 0.5074118 0.4956269 0.4880325 0.4826526 0.4786632 0.4756031
## [4,] 0.5339468 0.5095300 0.4967257 0.4888651 0.4834460 0.4794714 0.4764414
## [5,] 0.5581241 0.5479706 0.5418745 0.5374554 0.5339538 0.5310720 0.5286361
## [6,] 0.5911669 0.5669856 0.5537723 0.5452388 0.5390943 0.5344080 0.5307284
## [7,] 0.6172011 0.5833706 0.5648485 0.5526696 0.5439795 0.5374559 0.5323468
## [8,] 0.6278482 0.5873678 0.5673766 0.5548743 0.5461792 0.5397108 0.5346846
##           [,8]      [,9]     [,10]     [,11]     [,12]     [,13]     [,14]
## [1,] 0.4677740 0.4671630 0.4666495 0.4662162 0.4658545 0.4655551 0.4653090
## [2,] 0.4820703 0.4805587 0.4792811 0.4782188 0.4773077 0.4765522 0.4758984
## [3,] 0.4731856 0.4712544 0.4696852 0.4684039 0.4673551 0.4664921 0.4657894
## [4,] 0.4740724 0.4721534 0.4706145 0.4693499 0.4683016 0.4674367 0.4667275
## [5,] 0.5264965 0.5246474 0.5230199 0.5215688 0.5202826 0.5191147 0.5180880
## [6,] 0.5277346 0.5252756 0.5231759 0.5214098 0.5198813 0.5185995 0.5174582
## [7,] 0.5282643 0.5249055 0.5221242 0.5198020 0.5178234 0.5161232 0.5146520
## [8,] 0.5306878 0.5274150 0.5246848 0.5223845 0.5204266 0.5187525 0.5172813
##          [,15]     [,16]     [,17]     [,18]     [,19]     [,20]     [,21]
## [1,] 0.4651117 0.4649701 0.4648516 0.4647720 0.4647380 0.4647185 0.4647489
## [2,] 0.4753522 0.4748894 0.4745008 0.4741900 0.4739212 0.4737079 0.4735473
## [3,] 0.4652119 0.4647405 0.4643597 0.4640625 0.4638321 0.4636621 0.4635670
## [4,] 0.4661415 0.4656691 0.4652771 0.4649623 0.4647166 0.4645512 0.4644172
## [5,] 0.5171561 0.5163254 0.5155939 0.5149164 0.5143044 0.5137822 0.5132850
## [6,] 0.5164800 0.5156091 0.5148548 0.5142252 0.5136482 0.5131417 0.5126980
## [7,] 0.5133962 0.5122812 0.5113227 0.5104943 0.5097683 0.5091318 0.5085962
## [8,] 0.5160162 0.5149176 0.5139598 0.5131063 0.5123639 0.5117033 0.5111430
##          [,22]     [,23]     [,24]     [,25]     [,26]     [,27]     [,28]
## [1,] 0.4647866 0.4648599 0.4649683 0.4650642 0.4652059 0.4653634 0.4655179
## [2,] 0.4734177 0.4733380 0.4732907 0.4732753 0.4732900 0.4733282 0.4733887
## [3,] 0.4634951 0.4634749 0.4634996 0.4635494 0.4636239 0.4637493 0.4638743
## [4,] 0.4643285 0.4643042 0.4642938 0.4643401 0.4643928 0.4644874 0.4646048
## [5,] 0.5128430 0.5124876 0.5121406 0.5118469 0.5115899 0.5113784 0.5111924
## [6,] 0.5123391 0.5120283 0.5117522 0.5115247 0.5113479 0.5111933 0.5110901
## [7,] 0.5081215 0.5077090 0.5073407 0.5070733 0.5068017 0.5066042 0.5064413
## [8,] 0.5106477 0.5102100 0.5098400 0.5095091 0.5092395 0.5090150 0.5088168
##          [,29]     [,30]
## [1,] 0.4657064 0.4659112
## [2,] 0.4734757 0.4735907
## [3,] 0.4640418 0.4642090
## [4,] 0.4647519 0.4649165
## [5,] 0.5110241 0.5109066
## [6,] 0.5109968 0.5109500
```

```
## [7,] 0.5063072 0.5062197
## [8,] 0.5086679 0.5085461
##
## $SE_mat
##             [,1]        [,2]        [,3]        [,4]        [,5]        [,6]
## [1,] 0.002481530 0.002034663 0.001762670 0.001574728 0.001435606 0.001328633
## [2,] 0.006648094 0.006270975 0.006028331 0.005876875 0.005785159 0.005729735
## [3,] 0.011885148 0.010113851 0.009389805 0.009013220 0.008784421 0.008618638
## [4,] 0.008608393 0.008655937 0.008336879 0.008086794 0.007892298 0.007733632
## [5,] 0.009311561 0.009738147 0.009869029 0.009883291 0.009843499 0.009773842
## [6,] 0.009960737 0.007711461 0.006739494 0.006323542 0.006142161 0.006090279
## [7,] 0.010959725 0.009484337 0.008620160 0.008132688 0.007821018 0.007615415
## [8,] 0.007512805 0.007995269 0.008306397 0.008467280 0.008565743 0.008619078
##             [,7]        [,8]        [,9]       [,10]       [,11]       [,12]
## [1,] 0.001243960 0.001177616 0.001126738 0.001086754 0.001058782 0.001039205
## [2,] 0.005699678 0.005691909 0.005690094 0.005694747 0.005711973 0.005723667
## [3,] 0.008505790 0.008413204 0.008337574 0.008275626 0.008215255 0.008165627
## [4,] 0.007600993 0.007491539 0.007402221 0.007319938 0.007253853 0.007198222
## [5,] 0.009674098 0.009589990 0.009481450 0.009370637 0.009266891 0.009162205
## [6,] 0.006100369 0.006143886 0.006226476 0.006328353 0.006433955 0.006546431
## [7,] 0.007470574 0.007388722 0.007321900 0.007265999 0.007237733 0.007215214
## [8,] 0.008665674 0.008683204 0.008693634 0.008703996 0.008703964 0.008694646
##            [,13]       [,14]       [,15]       [,16]       [,17]       [,18]
## [1,] 0.001027663 0.001023183 0.001023717 0.001032738 0.001041009 0.001052623
## [2,] 0.005742200 0.005764682 0.005786605 0.005809620 0.005833824 0.005859257
## [3,] 0.008114939 0.008070176 0.008028200 0.007988614 0.007951119 0.007919158
## [4,] 0.007144588 0.007099332 0.007059857 0.007027290 0.006997246 0.006971034
## [5,] 0.009067083 0.008964653 0.008868955 0.008777787 0.008685545 0.008602086
## [6,] 0.006656987 0.006782915 0.006901756 0.007013535 0.007126913 0.007251017
## [7,] 0.007204637 0.007199051 0.007189880 0.007189361 0.007200919 0.007206837
## [8,] 0.008688902 0.008678436 0.008660563 0.008645025 0.008617069 0.008594437
##            [,19]       [,20]       [,21]       [,22]       [,23]       [,24]
## [1,] 0.001068819 0.001085685 0.001106756 0.001125545 0.001147237 0.001171472
## [2,] 0.005883939 0.005910072 0.005934891 0.005960753 0.005985732 0.006010120
## [3,] 0.007886079 0.007854891 0.007814850 0.007786035 0.007753828 0.007722954
## [4,] 0.006948912 0.006925626 0.006909206 0.006895309 0.006879011 0.006869224
## [5,] 0.008522573 0.008446531 0.008374703 0.008309439 0.008232692 0.008171658
## [6,] 0.007366496 0.007480212 0.007592014 0.007708955 0.007803283 0.007913160
## [7,] 0.007214227 0.007222734 0.007227221 0.007243352 0.007248028 0.007258784
## [8,] 0.008575422 0.008551036 0.008524586 0.008496717 0.008475659 0.008438243
##            [,25]       [,26]       [,27]       [,28]       [,29]       [,30]
## [1,] 0.001192100 0.001217863 0.001240193 0.001264607 0.001286581 0.001312978
## [2,] 0.006034571 0.006059607 0.006083879 0.006107763 0.006132107 0.006154285
## [3,] 0.007692826 0.007672638 0.007639781 0.007618622 0.007586723 0.007567439
## [4,] 0.006856484 0.006850074 0.006845094 0.006834082 0.006829040 0.006828633
## [5,] 0.008110080 0.008049159 0.007993430 0.007943961 0.007887685 0.007843584
## [6,] 0.008024325 0.008111502 0.008218394 0.008306386 0.008409245 0.008493178
## [7,] 0.007280110 0.007291274 0.007291700 0.007319347 0.007326006 0.007347921
## [8,] 0.008414195 0.008387376 0.008362844 0.008330933 0.008306631 0.008268928
##
## $opt_para
##      [,1] [,2] [,3]
## [1,]    5    2   23
```

Check explained variability

```r
opt_res <- mglram(tnsr = synthetic_dat@data, ranks = c(5, 2), init=0, D = D2,
       lambda = 23, max_iter = 500, tol = 1e-5, L0 = NULL)
```

```r
# Cumulative explained variability when r1 increases
tilde <- MakeIdent(opt_res$L, opt_res$G, opt_res$R)
L_tilde <- tilde$L_tilde
G_tilde <- tilde$G_tilde
R_tilde <- tilde$R_tilde

nmiss_idx <- which(!is.na(synthetic_dat@data))

cum_var_L <- rep(NA, 5)
for (i in 1:5){
  comp <- array(NA, c(24,3,207))
  for (j in 1:207){
    comp[ , , j] <- matrix(L_tilde[, 1:i], ncol=i) %*% matrix(G_tilde[1:i, , j], nrow=i) %*% t(R_tilde)
  }
  cum_var_L[i] <- sum(comp[nmiss_idx]^2)/sum(synthetic_dat@data[nmiss_idx]^2)
}

cum_var_L # 0.4448095 0.5574384 0.6032486 0.6055275 0.6065202
```

```
## [1] 0.4448095 0.5574384 0.6032486 0.6055275 0.6065202
```

```r
# Separate explained variability for different L ranks
cum_var_L0 <- c(0, cum_var_L[1:4])
sep_var_L <- cum_var_L - cum_var_L0

sep_var_L # 0.4448094949 0.1126289400 0.0458101564 0.0022789411 0.0009927149
```

```
## [1] 0.4448094949 0.1126289400 0.0458101564 0.0022789411 0.0009927149
```

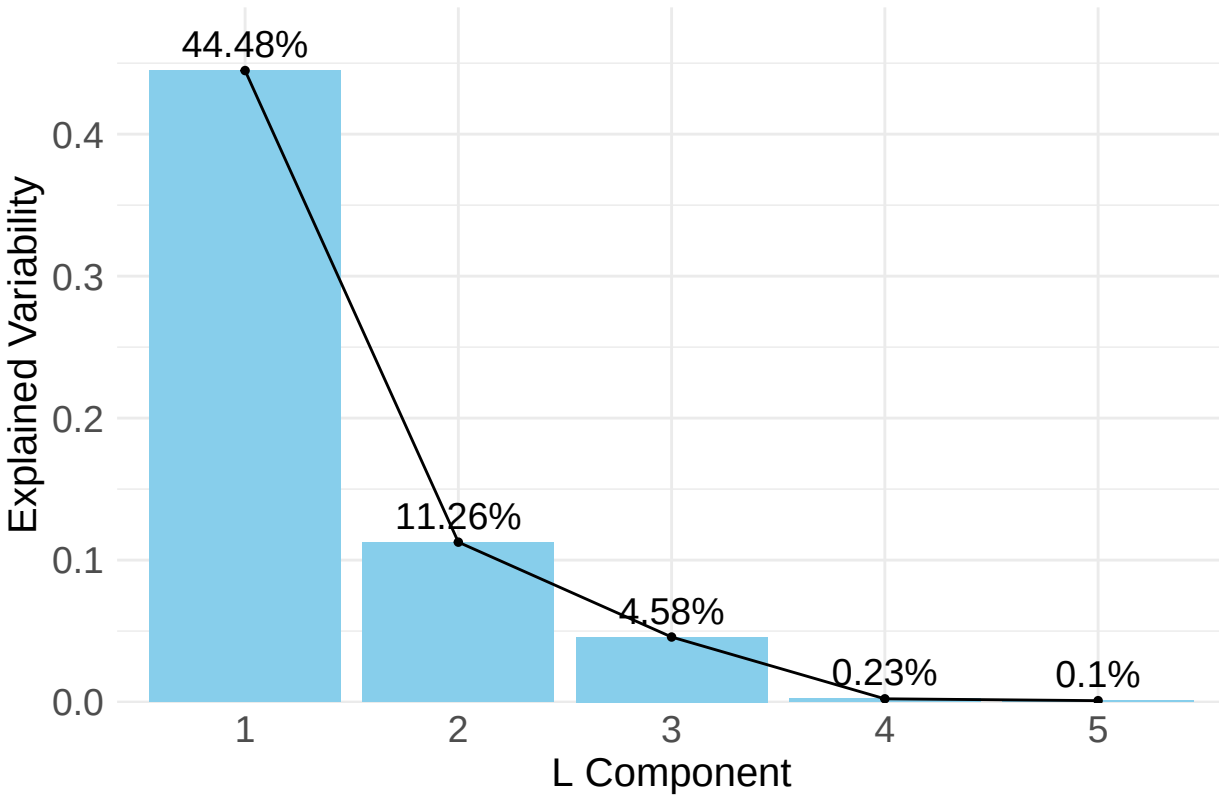
```r
# Cumulative explained variability when r2 increases
cum_var_R <- rep(NA, 2)
for (i in 1:2){
  comp <- array(NA, dim(synthetic_dat@data))
  for (j in 1:207){
    comp[ , , j] <- L_tilde %*% matrix(G_tilde[,1:i, j], ncol=i) %*% matrix(t(R_tilde[, 1:i]), nrow=i)
  }
  cum_var_R[i] <- sum(comp[nmiss_idx]^2)/sum(synthetic_dat@data[nmiss_idx]^2)
}

cum_var_R #  0.3966868 0.6065202
```

```
## [1] 0.3966868 0.6065202
```

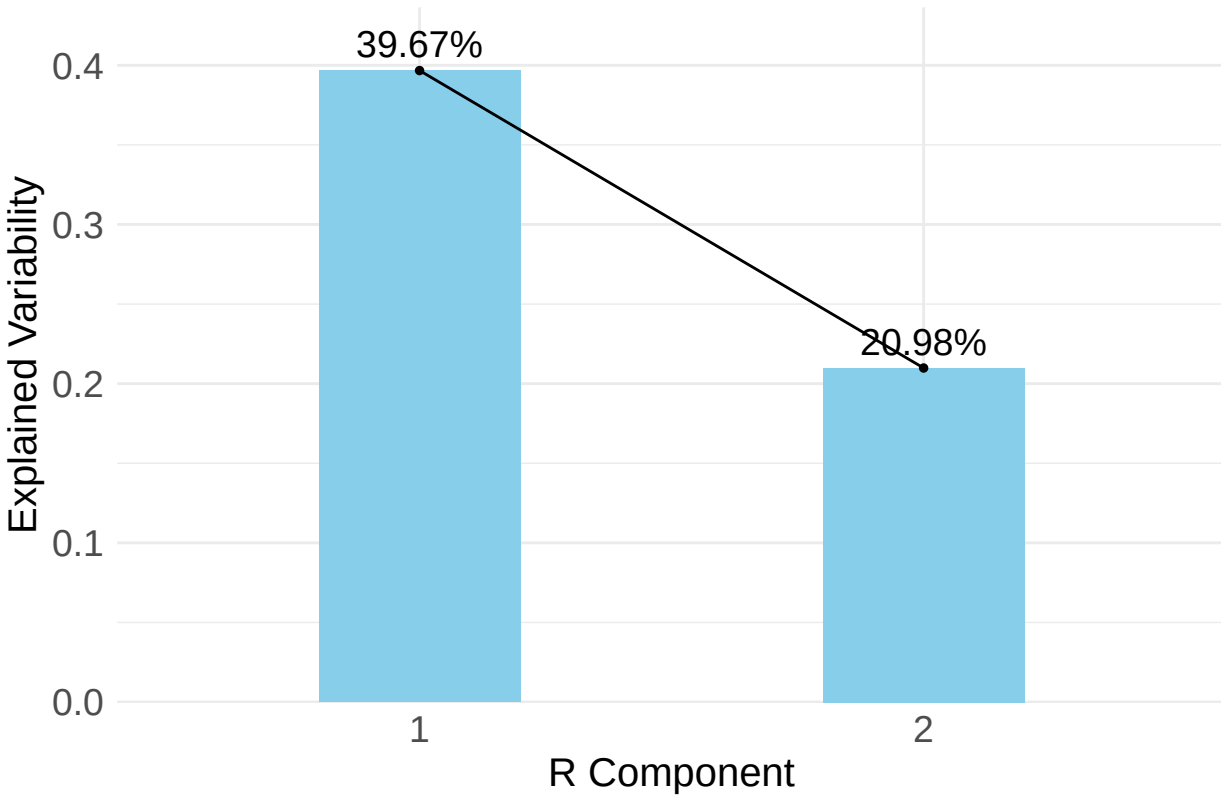
```r
# Separate explained variability for different R ranks
cum_var_R0 <- c(0, cum_var_R[1:1])
sep_var_R <- cum_var_R - cum_var_R0

sep_var_R # 0.3966868 0.2098334
```

```
## [1] 0.3966868 0.2098334
```

```r
# Plot the explained variability
var_data_L <- data.frame(
  component = factor(seq_along(sep_var_L), levels = seq_along(sep_var_L)),
  value = sort(sep_var_L, decreasing = TRUE)
)

var_data_L$percentage <- paste0(round(var_data_L$value * 100, 2), "%")

ggplot(var_data_L, aes(x = component, y = value)) +
  geom_bar(stat = "identity", fill = "skyblue") +
  geom_text(aes(label = percentage), vjust = -0.5, size = 5) +
  geom_line(aes(group = 1), color = "black", linewidth = 0.5) +
  geom_point(color = "black", size = 1) +
  labs(y = "Explained Variability", x="L Component", title="") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.title.y = element_text(size = 15),
        axis.title.x = element_text(size = 15),
        axis.text.x = element_text(size = 14),
        axis.text.y = element_text(size = 14)) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1)))
```




```r
# Plot the explained variability
var_data_R <- data.frame(
  component = factor(seq_along(sep_var_R), levels = seq_along(sep_var_R)),
  value = sort(sep_var_R, decreasing = TRUE)
)

var_data_R$percentage <- paste0(round(var_data_R$value * 100, 2), "%")

ggplot(var_data_R, aes(x = component, y = value)) +
  geom_bar(stat = "identity", fill = "skyblue", width = 0.4) +
  geom_text(aes(label = percentage), vjust = -0.5, size = 5) +
  geom_line(aes(group = 1), color = "black", linewidth = 0.5) +
  geom_point(color = "black", size = 1) +
  labs(title = "", x = "R Component", y = "Explained Variability") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.title.y = element_text(size = 15),
        axis.title.x = element_text(size = 15),
        axis.text.x = element_text(size = 14),
        axis.text.y = element_text(size = 14)) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1)))
```



Update hyperparameters for parsimony

```
start_time <- Sys.time()

res <- mglram(tnsr = synthetic_dat@data, ranks = c(3, 2), init=0, D = D2,
       lambda = 20, max_iter = 500, tol = 1e-5, L0 = NULL)

res$conv
```

```
## [1] TRUE
```

```
end_time <- Sys.time()

exec_time <- end_time - start_time
print(exec_time)
```

```
## Time difference of 0.465204 secs
```

```
tilde <- MakeIdent(L=res$L, G=res$G, R=res$R)

L_tilde <- tilde$L_tilde
R_tilde <- tilde$R_tilde
G_tilde <- tilde$G_tilde

L_tilde
```

```
##              [,1]        [,2]         [,3]
##  [1,] -0.2249415  0.17526918 -0.192070913
##  [2,] -0.2253011  0.21690826 -0.131473011
##  [3,] -0.2246080  0.24470898 -0.063553498
##  [4,] -0.2230082  0.25746373  0.006939944
##  [5,] -0.2205265  0.25450393  0.075500664
##  [6,] -0.2170161  0.23554347  0.138545506
##  [7,] -0.2124737  0.20056942  0.193529241
##  [8,] -0.2068758  0.15026238  0.237983437
##  [9,] -0.2004723  0.08702414  0.269071918
## [10,] -0.1936527  0.01408099  0.284840755
## [11,] -0.1871342 -0.06388877  0.284591811
## [12,] -0.1816032 -0.14086089  0.267130440
## [13,] -0.1778034 -0.21012392  0.231996680
## [14,] -0.1761792 -0.26573484  0.179079552
## [15,] -0.1770223 -0.30287804  0.111765084
## [16,] -0.1801498 -0.31760917  0.033569768
## [17,] -0.1851735 -0.30829276 -0.049040390
## [18,] -0.1916734 -0.27631595 -0.128641601
## [19,] -0.1986725 -0.22528080 -0.198279904
## [20,] -0.2054921 -0.16043814 -0.250555867
## [21,] -0.2115149 -0.08766743 -0.281364343
## [22,] -0.2164138 -0.01326350 -0.289041914
## [23,] -0.2202998  0.05798688 -0.274740554
## [24,] -0.2232133  0.12172947 -0.241072190
```

```
R_tilde
```

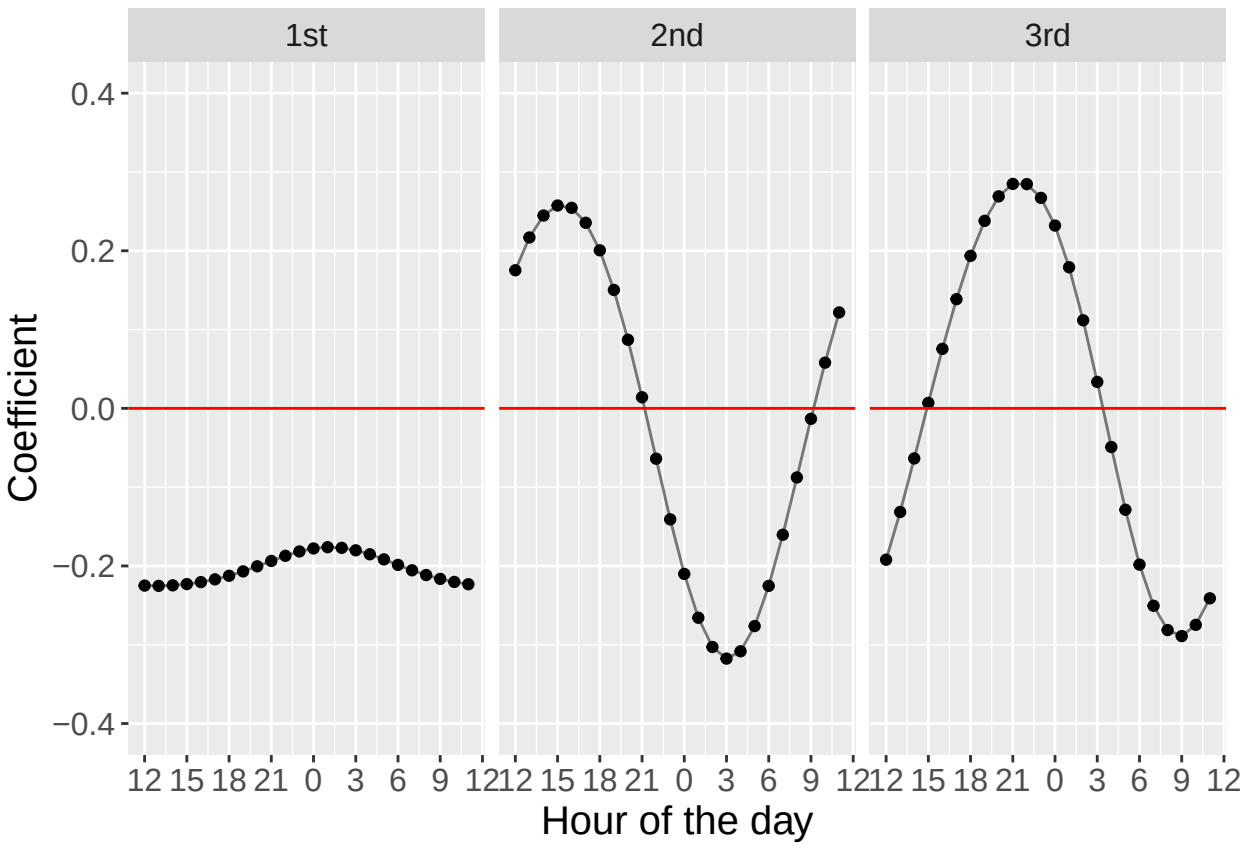
```
##            [,1]       [,2]
## [1,] -0.6613123  0.1629765
## [2,] -0.6044204  0.4622982
## [3,] -0.4442319 -0.8716186
```

```
G_tilde[ , ,1:5]
```

```
## , , 1
##
##           [,1]       [,2]
## [1,] -4.225260 -1.2150099
## [2,] -1.037436  0.7843903
## [3,] -3.027923 -0.6656535
##
## , , 2
##
##             [,1]       [,2]
## [1,] -0.99176614  1.3842688
## [2,] -1.46806739 -0.1779750
## [3,]  0.03566204  0.1016606
##
## , , 3
##
##            [,1]       [,2]
## [1,] -2.5677201 -0.3949555
```

```
## [2,] -1.7291657  0.5253079
## [3,]  0.3166862 -0.1719826
##
## , , 4
##
##            [,1]        [,2]
## [1,] -2.4971277 -7.61477112
## [2,] -1.9615627  0.26540029
## [3,] -0.7520841 -0.05225598
##
## , , 5
##
##           [,1]       [,2]
## [1,] -3.848932  2.5746534
## [2,] -2.023032  0.1151673
## [3,]  0.483573 -0.8019558
```

```
dataL = data.frame(loading = c(L_tilde[, 1], L_tilde[, 2], L_tilde[, 3]),
                   component = c(rep("1st", 24), rep("2nd", 24), rep("3rd", 24)),
                   hour = rep(12:35, 3))

pL = dataL %>%
  ggplot(aes(x = hour, y = loading)) + geom_line(alpha = 0.5) + geom_point() + geom_hline(yintercept = 0
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6
  xlab("Hour of the day") +
  facet_grid(~ component) +
  ylab("Coefficient") +
  theme(text = element_text(size = 15)) +
  scale_y_continuous(limits = c(-0.4, 0.4))

pL
```




```
R_tilde[, 1] # first component of R tilde
```

```
## [1] -0.6613123 -0.6044204 -0.4442319
```

```
R_tilde[, 2] # second component of R tilde
```

```
## [1]  0.1629765  0.4622982 -0.8716186
```

```
## 36/100 DBP + 31/50 SBP + 33/100 HR (Seem to be joint effect of the three)
R_tilde[1, 1]/sum(R_tilde[, 1]) # DBP: 39/100
```

```
## [1] 0.3867403
```

```
R_tilde[2, 1]/sum(R_tilde[, 1]) # SBP: 35/100
```

```
## [1] 0.3534695
```

```
R_tilde[3, 1]/sum(R_tilde[, 1]) # HR: 26/100
```

```
## [1] 0.2597901
```

```r
## 17/100 DBP + 33/100 SBP - 50/100 HR (Contrast between blood pressure and heart rate)
R_tilde[1, 2]/sum(abs(R_tilde[, 2])) # DBP: 11/100
```

```
## [1] 0.1088765
```

```r
R_tilde[2, 2]/sum(abs(R_tilde[, 2])) # SBP: 31/100
```

```
## [1] 0.3088385
```

```r
R_tilde[3, 2]/sum(abs(R_tilde[, 2])) # HR: -58/100
```

```
## [1] -0.5822851
```

```r
# Prepare to plot effect of time components
n = 207

Ri = array(NA, dim = c(3, 3, n))
for (i in 1:n){
  Ri[, , i] = G_tilde[, , i] %*% t(R_tilde)
}

RiL1 = Ri[1, , ]
apply(RiL1, 1, sd) # 2.789581 2.997533 3.661661 (sd for DBP, SBP, and HR for 1st time component)
```

```
## [1] 2.897844 3.058188 3.579392
```

```r
RiL2 = Ri[2, , ]
apply(RiL2, 1, sd) # 1.379921 1.401255 1.444989 (sd for DBP, SBP, and HR for 2nd time component)
```

```
## [1] 1.310837 1.332944 1.365773
```

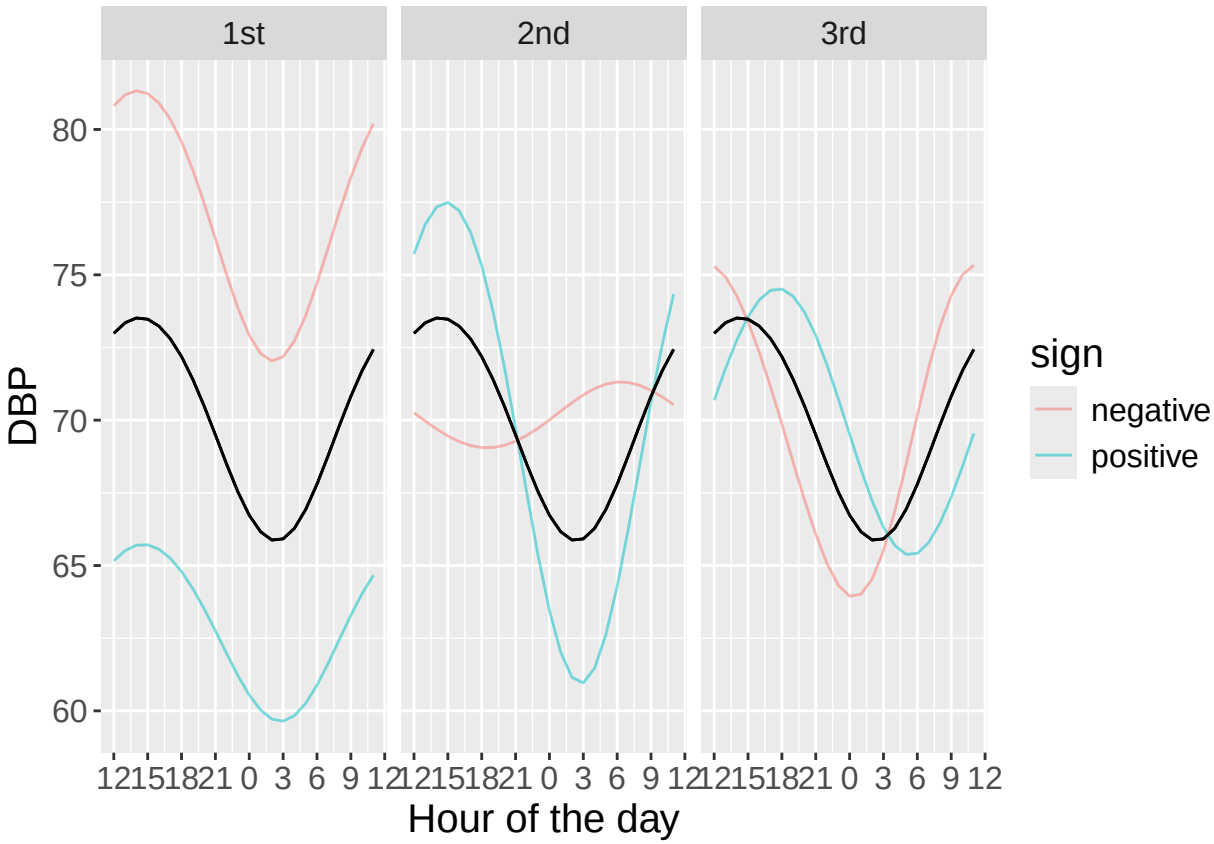
```r
RiL3 = Ri[3, , ]
apply(RiL3, 1, sd) # 1.067016 1.115904 1.072485 (sd for DBP, SBP, and HR for 3rd time component)
```

```
## [1] 0.9579192 1.0089143 1.0671482
```

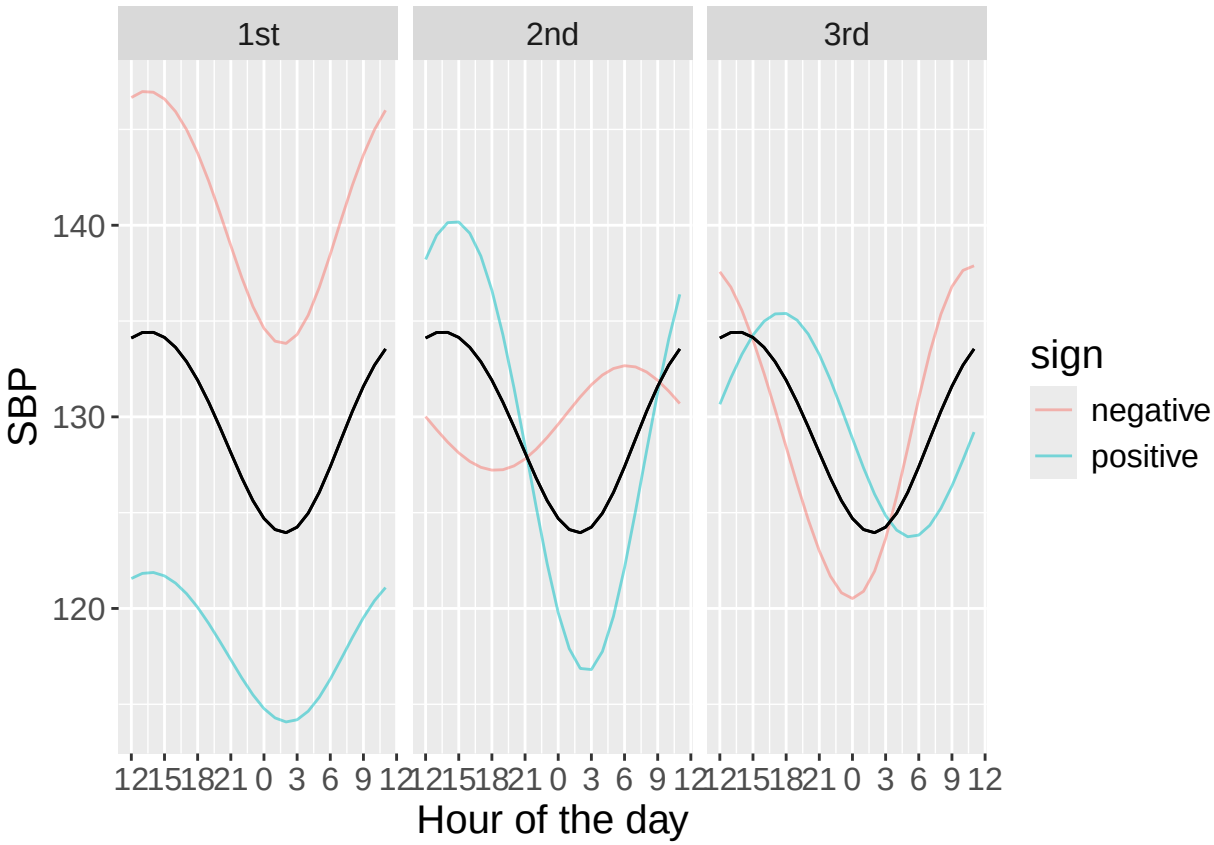
```r
DBPmean = rep(70, 24) + (L_tilde[, 1] * mean(RiL1[1,]) + L_tilde[, 2] * mean(RiL2[1,]) + L_tilde[, 3] *
SBPmean = rep(129, 24) + (L_tilde[, 1] * mean(RiL1[2,]) + L_tilde[, 2] * mean(RiL2[2,]) + L_tilde[, 3] *
HRmean = rep(76, 24) + (L_tilde[, 1] * mean(RiL1[3,]) + L_tilde[, 2] * mean(RiL2[3,]) + L_tilde[, 3] * r
```

```r
# Plot effect of time components on DBP
dataLeffects = data.frame(DBPmean = rep(DBPmean, 6),
                          value = c(DBPmean + L_tilde[, 1] * 2.9*12, DBPmean - L_tilde[, 1] * 2.9*12,
                                    DBPmean + L_tilde[, 2] *1.3*12, DBPmean - L_tilde[, 2] * 1.3*12,
                                    DBPmean + L_tilde[, 3] * 1.0*12, DBPmean - L_tilde[, 3] * 1.0*12),
                          component = c(rep("1st", 24 * 2), rep("2nd", 24 * 2), rep("3rd", 24 * 2)),
                          sign = rep(rep(c("positive", "negative"), each = 24), 3),
                          hour = rep(12:35, 6))
```

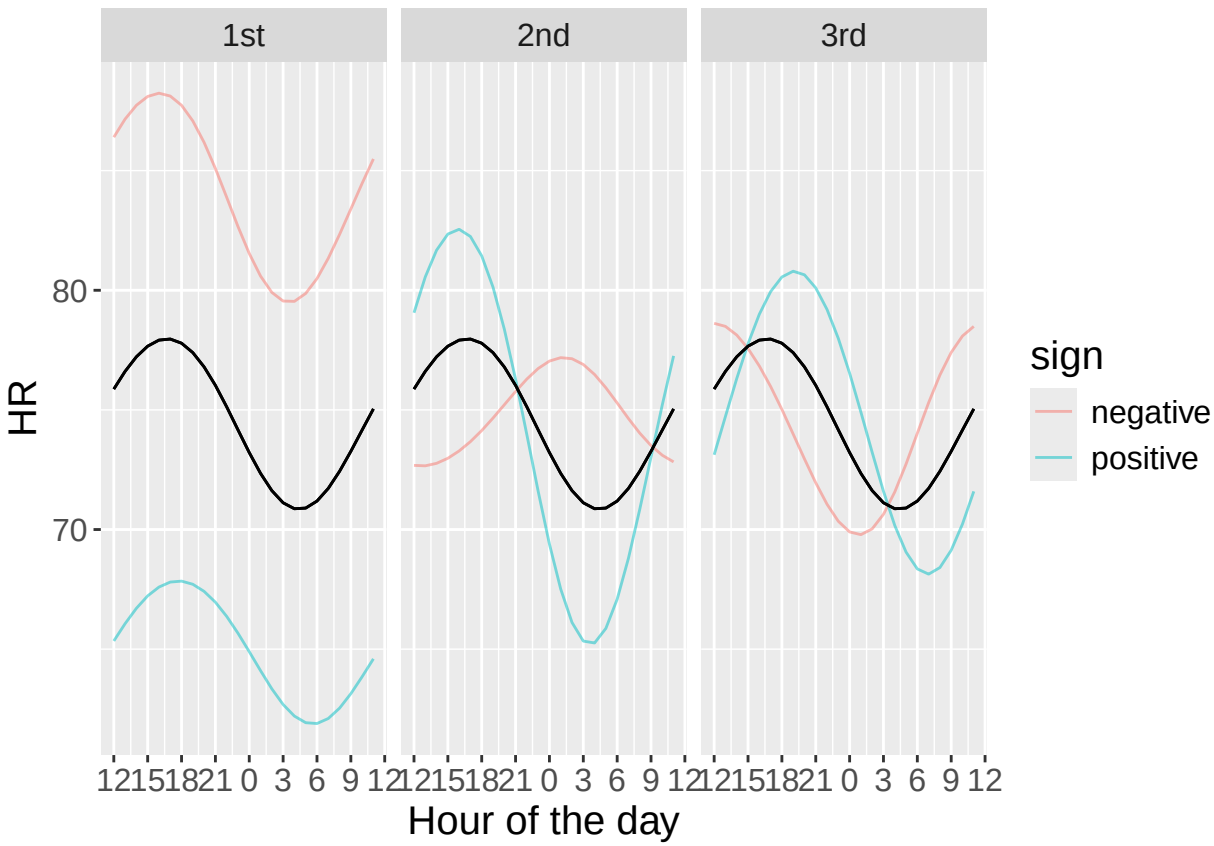
```
pdataL = dataLeffects %>%
  ggplot(aes(x = hour, y = value, group = sign, col = sign)) + geom_line(alpha = 0.5) +
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6
  xlab("Hour of the day") +
  facet_grid(~ component) +
  ylab("DBP") + geom_line(aes(x = hour, y = DBPmean), col = "black") +
  theme(text = element_text(size = 15))

pdataL
```




```
# Plot effect of time components on SBP
dataLeffects = data.frame(SBPmean = rep(SBPmean, 6),
                          value = c(SBPmean + L_tilde[, 1] * 3.1*18, SBPmean - L_tilde[, 1] * 3.1*18,
                                    SBPmean + L_tilde[, 2] * 1.3*18, SBPmean - L_tilde[, 2] * 1.3*18,
                                    SBPmean + L_tilde[, 3] * 1.0*18, SBPmean - L_tilde[, 3] * 1.0*18),
                          component = c(rep("1st", 24 * 2), rep("2nd", 24 * 2), rep("3rd", 24 * 2)),
                          sign = rep(rep(c("positive", "negative"), each = 24), 3),
                          hour = rep(12:35, 6))

pdataL = dataLeffects %>%
  ggplot(aes(x = hour, y = value, group = sign, col = sign)) + geom_line(alpha = 0.5) +
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6
  xlab("Hour of the day") + facet_grid(~ component) + ylab("SBP") + geom_line(aes(x = hour, y = SBPmean
  theme(text = element_text(size = 15))

pdataL
```

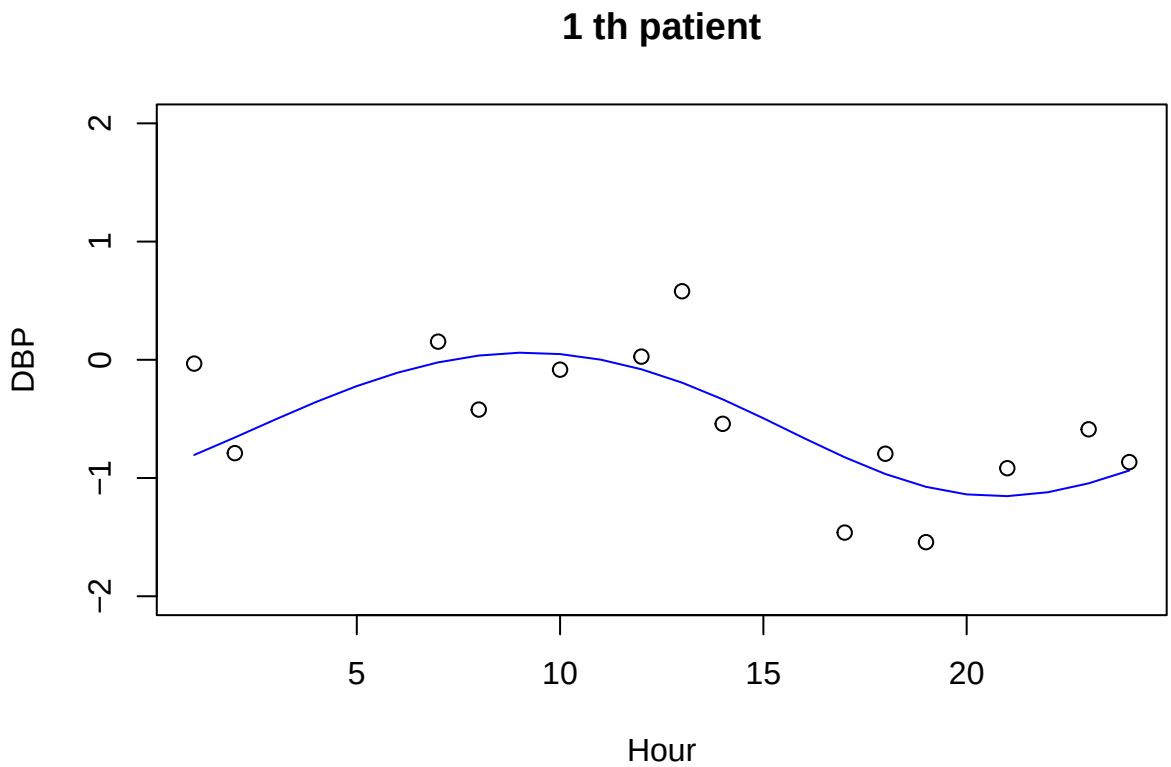


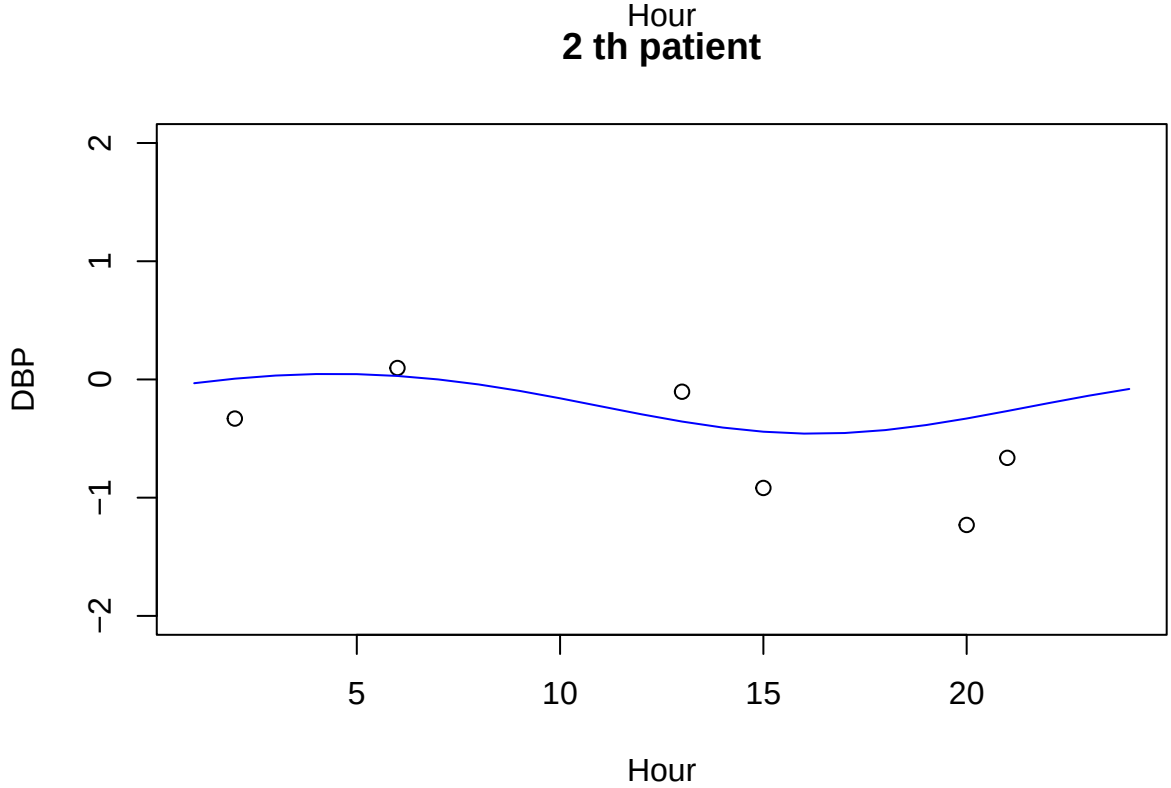

```
# Plot effect of time components on HR
dataLeffects = data.frame(HRmean = rep(HRmean, 6),
                          value = c(HRmean + L_tilde[, 1] * 3.6*13, HRmean - L_tilde[, 1] *3.6*13,
                                    HRmean + L_tilde[, 2] * 1.4*13, HRmean - L_tilde[, 2] * 1.4*13,
                                    HRmean + L_tilde[, 3] * 1.1*13, HRmean - L_tilde[, 3] * 1.1*13),
                          component = c(rep("1st", 24 * 2), rep("2nd", 24 * 2), rep("3rd", 24 * 2)),
                          sign = rep(rep(c("positive", "negative"), each = 24), 3),
                          hour = rep(12:35, 6))

pdataL = dataLeffects %>%
  ggplot(aes(x = hour, y = value, group = sign, col = sign)) + geom_line(alpha = 0.5) +
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6
  xlab("Hour of the day") + facet_grid(~ component) + ylab("HR") + geom_line(aes(x = hour, y = HRmean),
  theme(text = element_text(size = 15))

pdataL
```

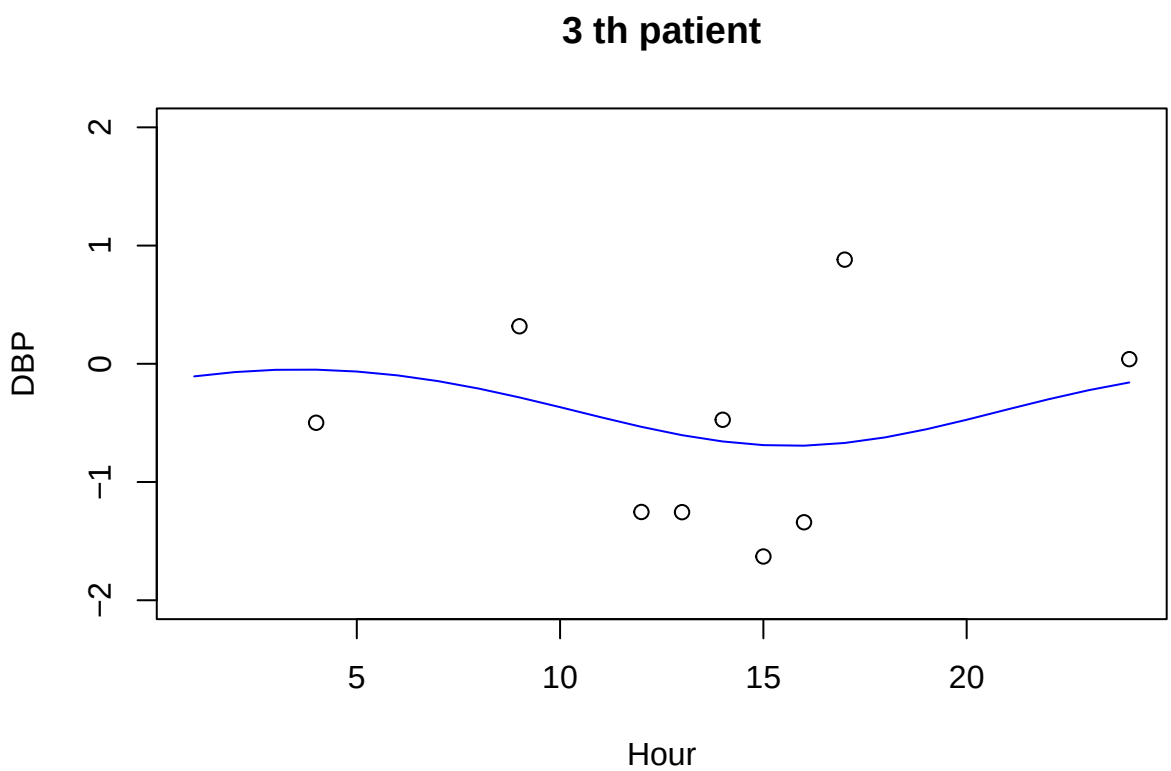


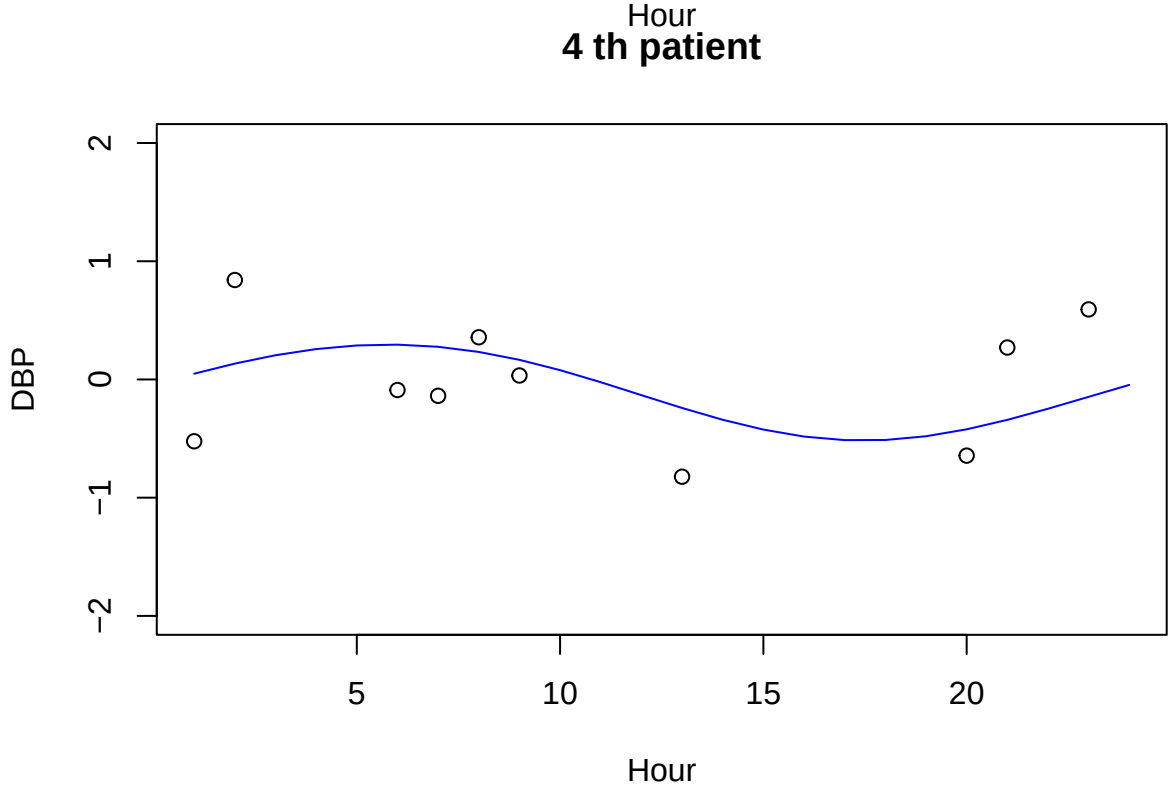
Plot estimated curves for some patients

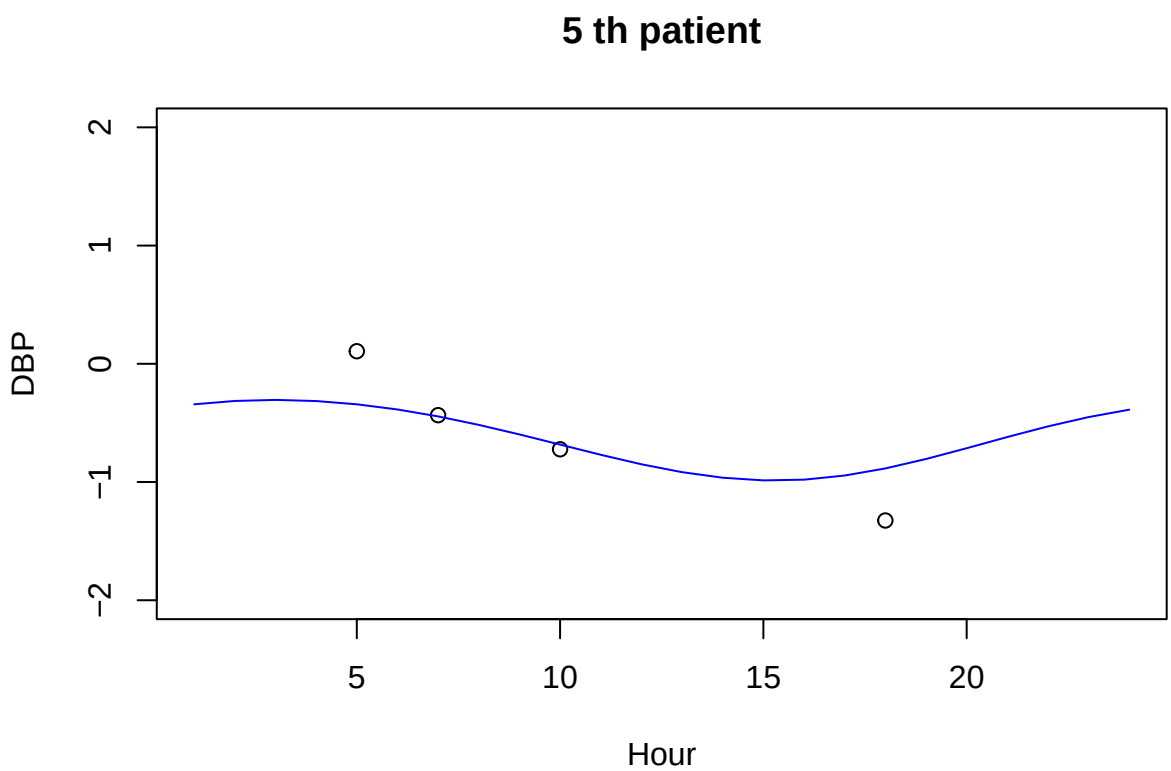
```
for (i in 1:5){
  plot(synthetic_dat@data[ , 1, i], main=paste(i,"th patient"), xlab="Hour", ylab="DBP", ylim=c(-2,2)) 
  lines(res$est[ , 1, i], col="blue") # estimated curve by our algorithm
}
```


1 th patient
DBP
2
1
0
−1
−2
5
10
15
20
Hour

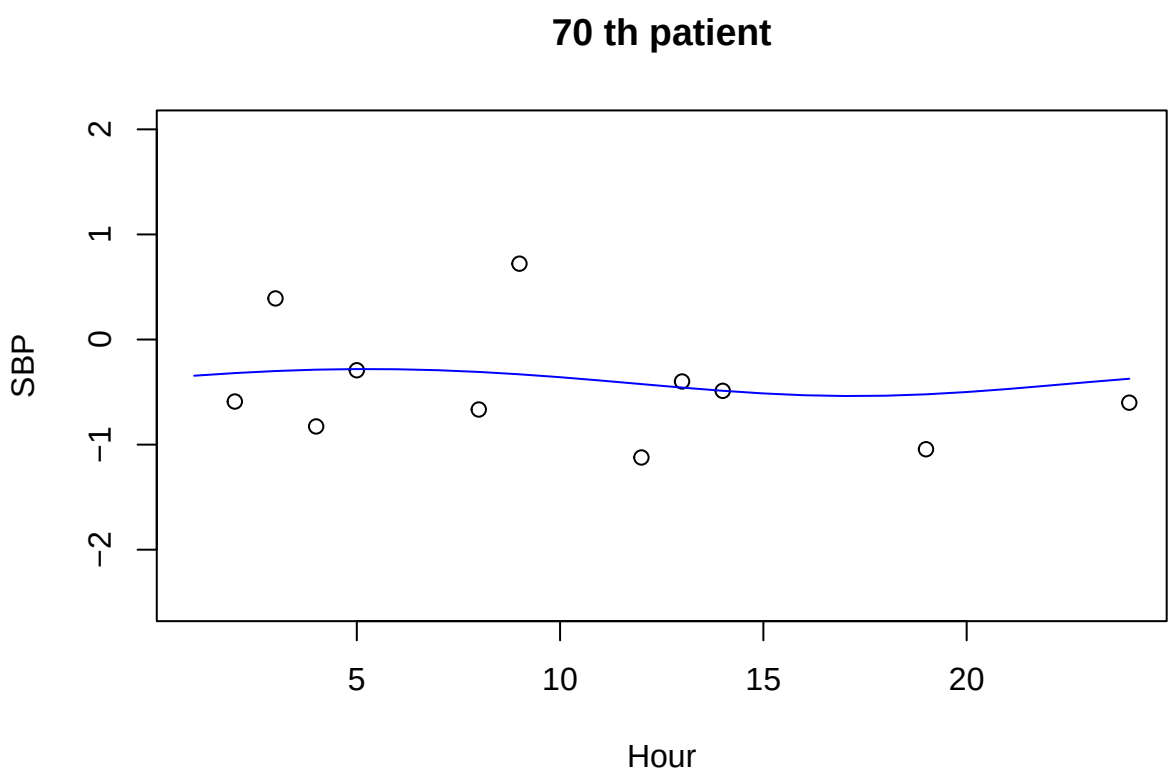

2 th patient
DBP
2
1
0
−1
−2
5
10
15
20
Hour

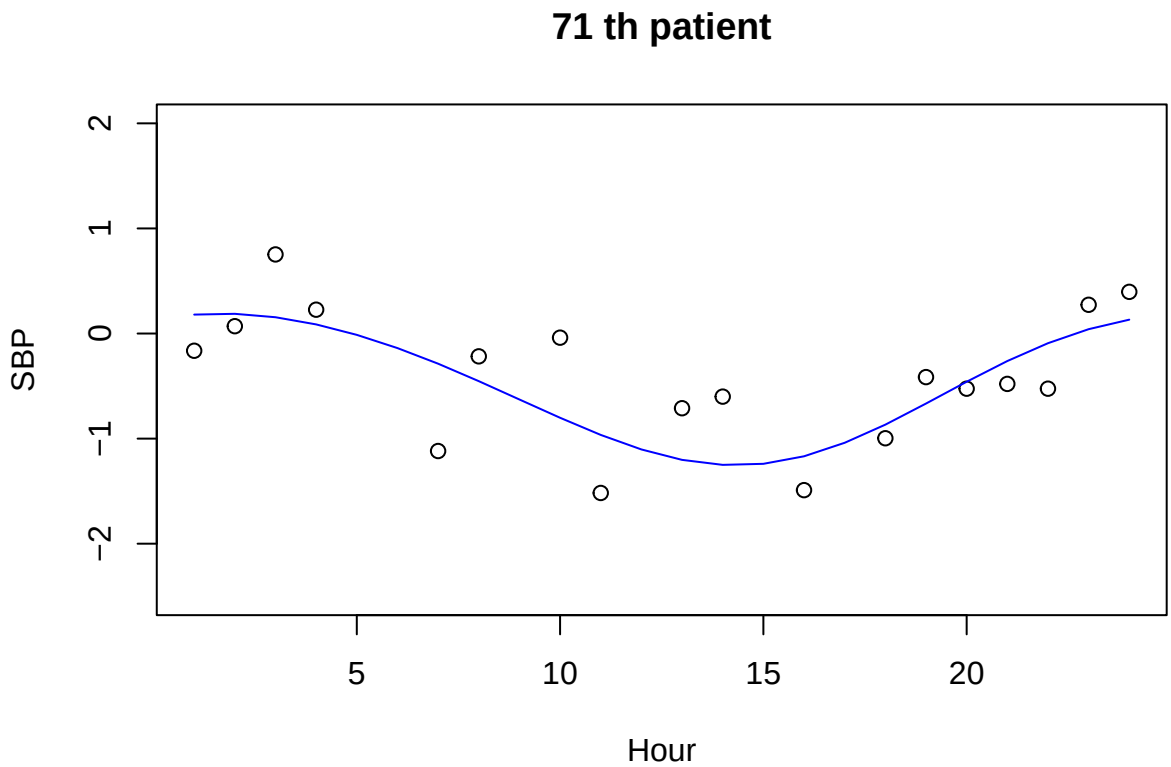

3 th patient
DBP
Hour
2
1
0
−1
−2
5
10
15
20

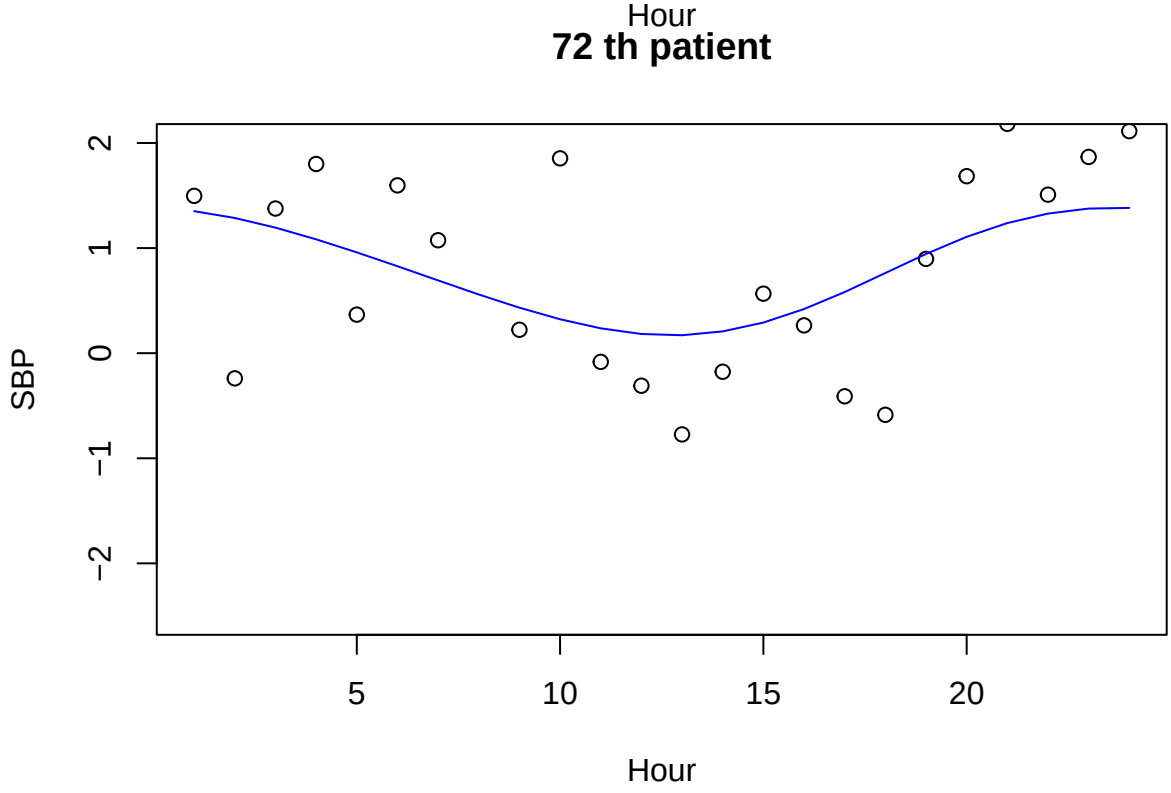

4 th patient
DBP
Hour
2
1
0
−1
−2
5
10
15
20

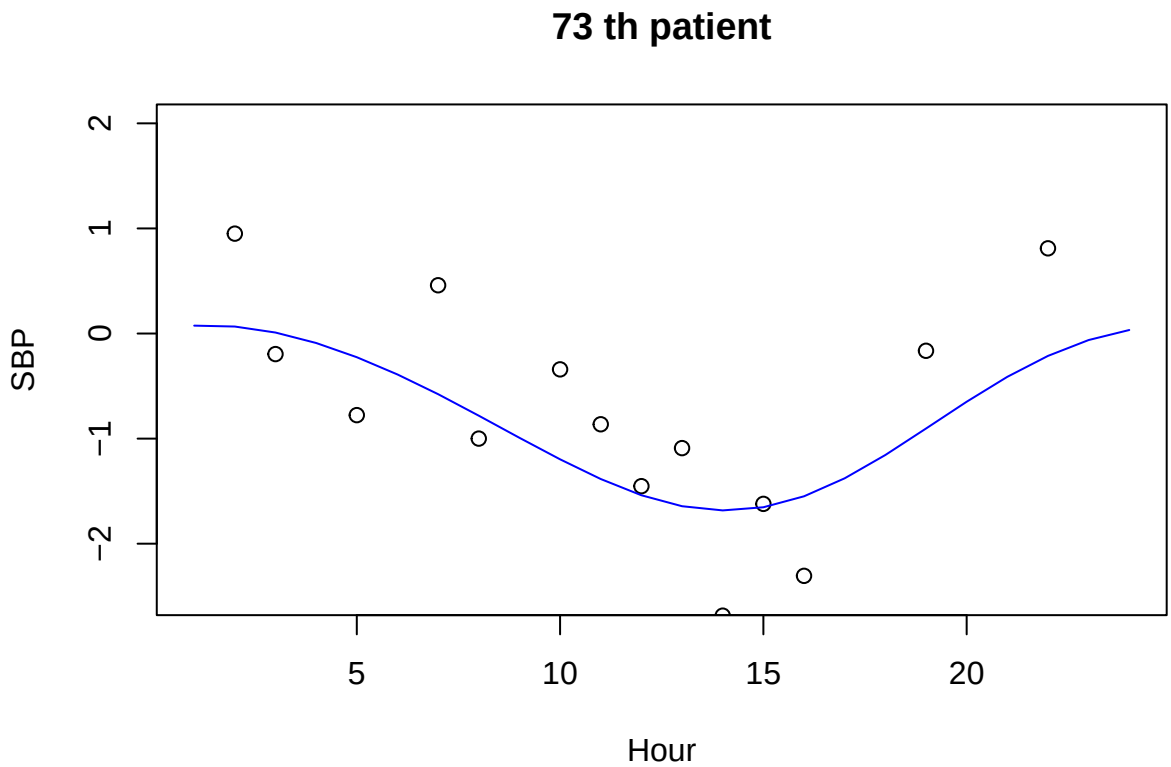


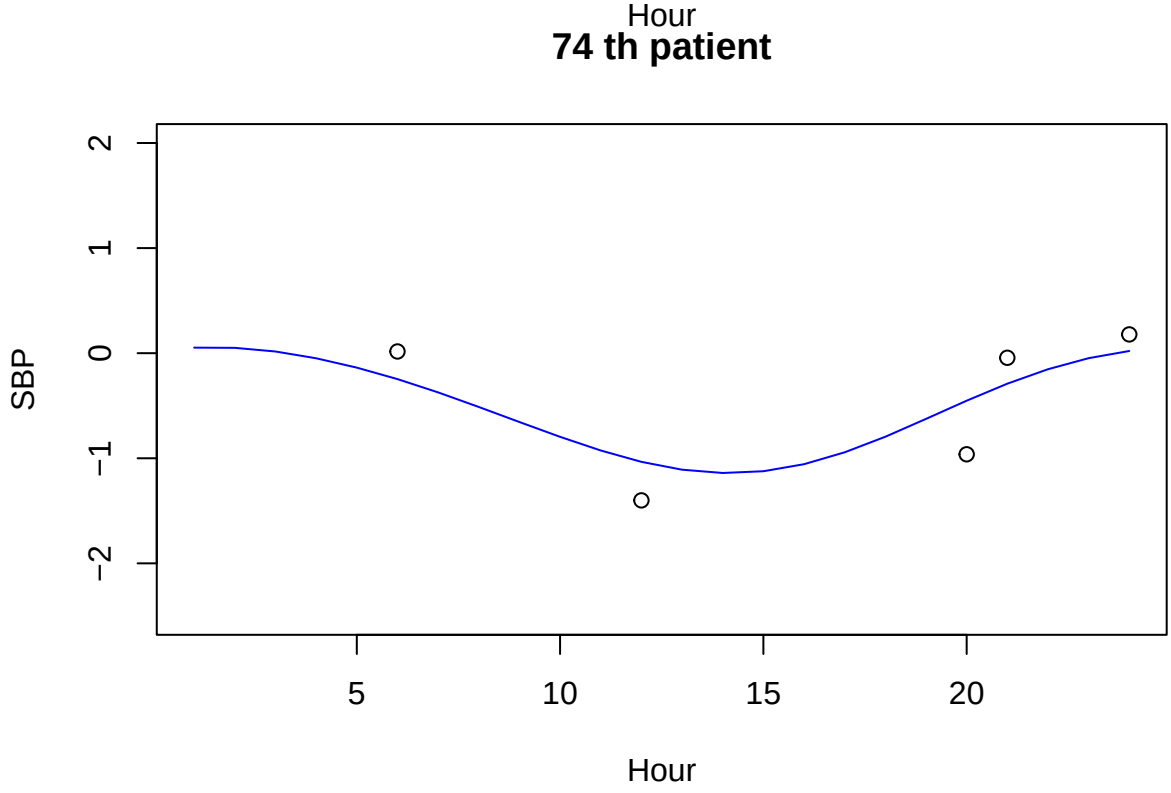

```
# Plot Estimated Curves for SBP
for (i in 70:75){
  plot(synthetic_dat@data[ , 2, i], main=paste(i,"th patient"), xlab="Hour", ylab="SBP", ylim=c(-2.5,2))
  lines(res$est[ , 2, i], col="blue")
}
```

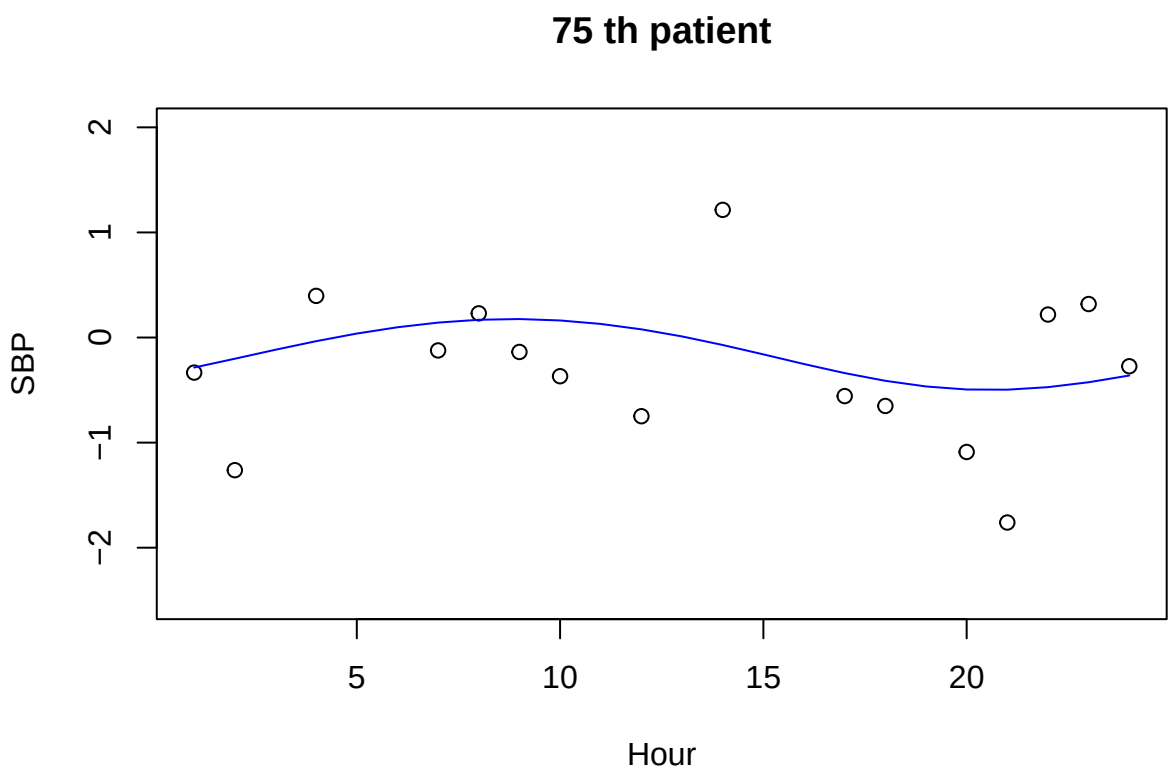



71 th patient
SBP
2
1
0
−1
−2
5
10
15
20
Hour

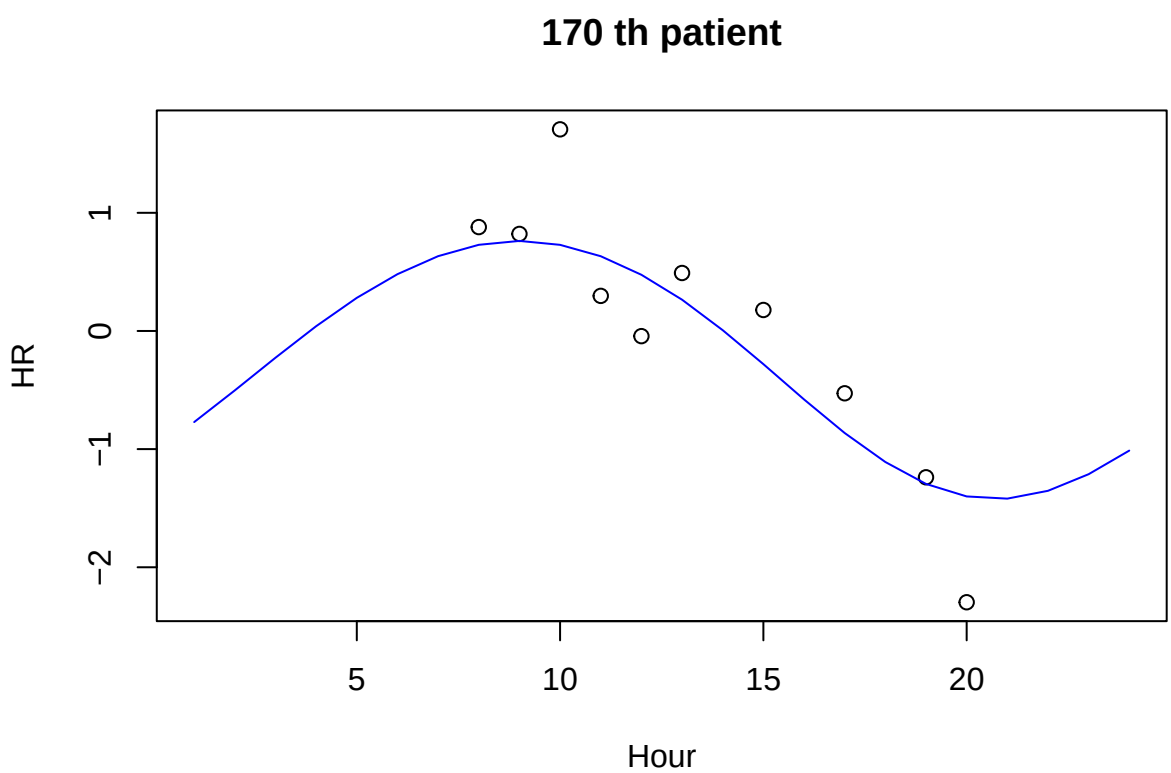

72 th patient
SBP
2
1
0
−1
−2
5
10
15
20
Hour

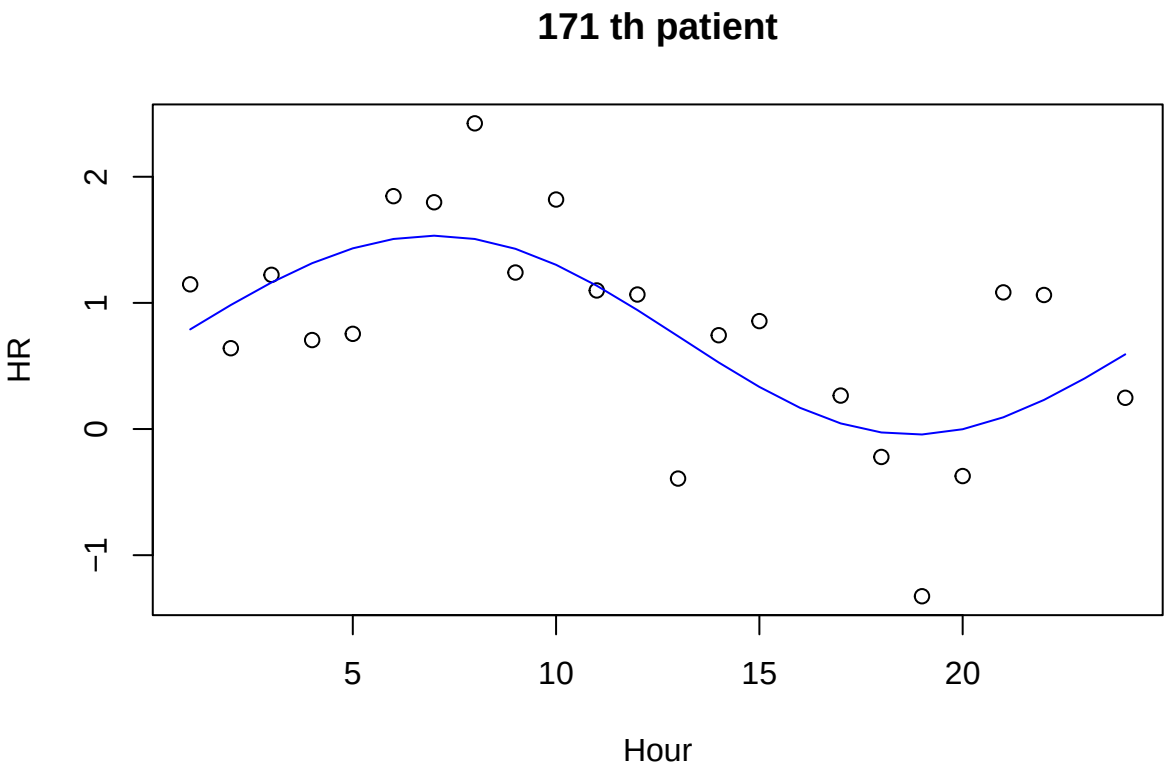

73 th patient
SBP
2
1
0
−1
−2
5
10
15
20
Hour

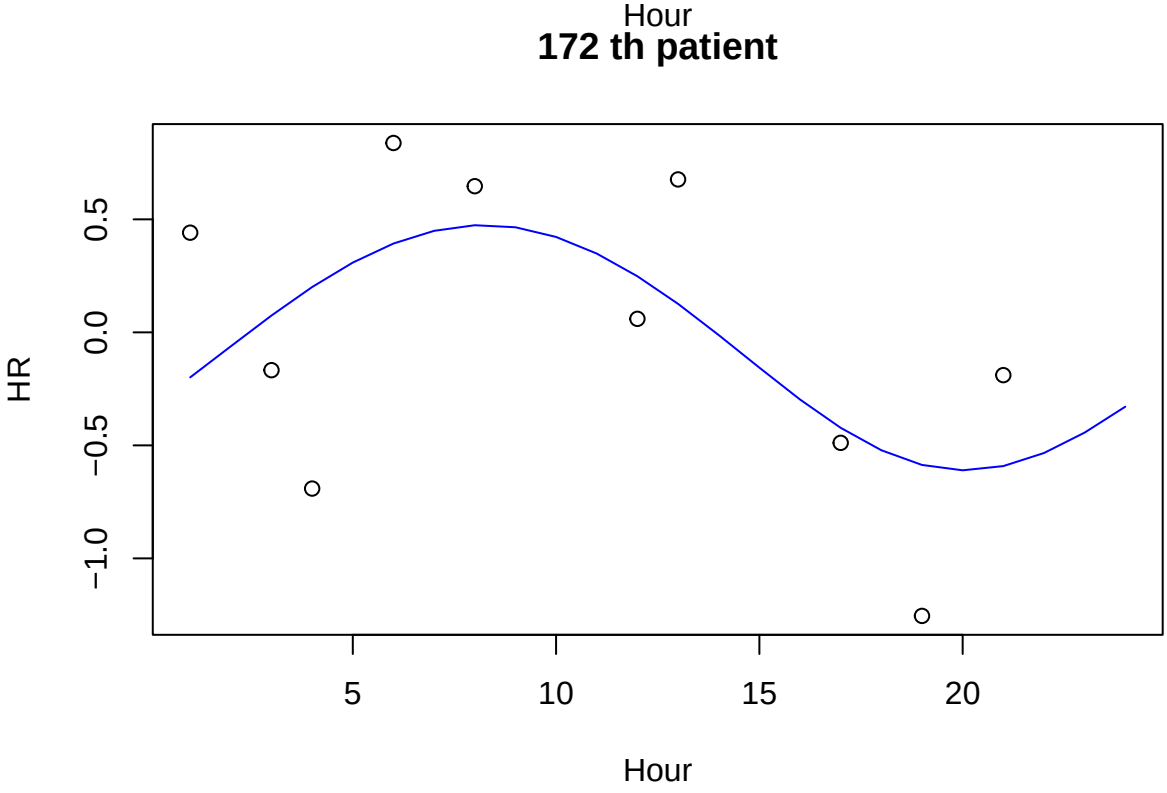

74 th patient
SBP
2
1
0
−1
−2
5
10
15
20
Hour

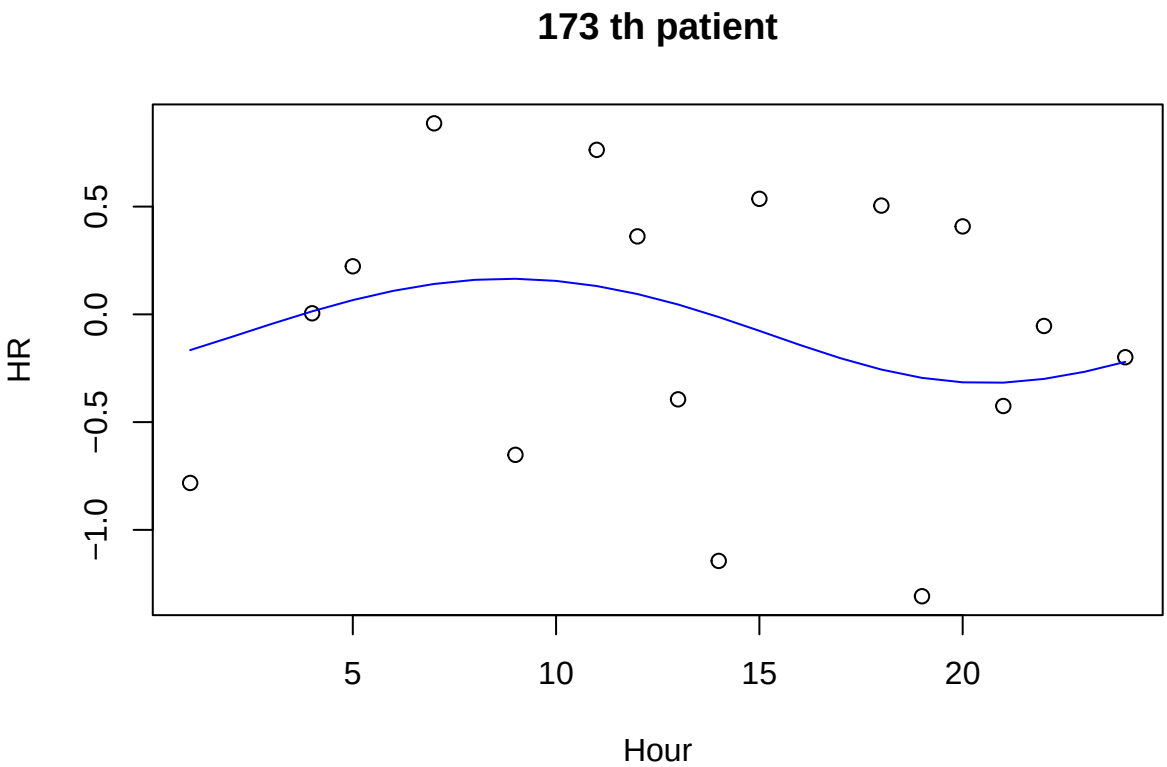


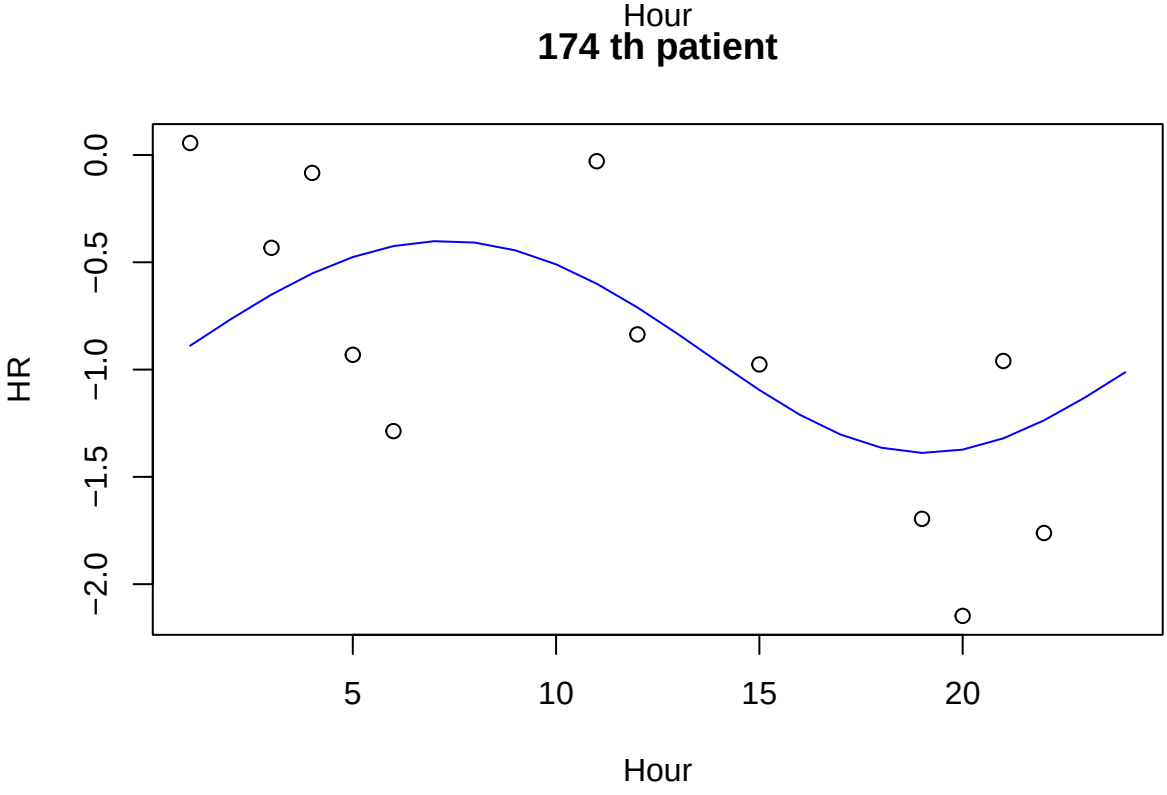

```r
# Plot Estimated Curves for HR
for (i in 170:175){
  plot(synthetic_dat@data[ , 3, i], main=paste(i,"th patient"), xlab="Hour", ylab="HR")
  lines(res$est[ , 3, i], col="blue")
}
```

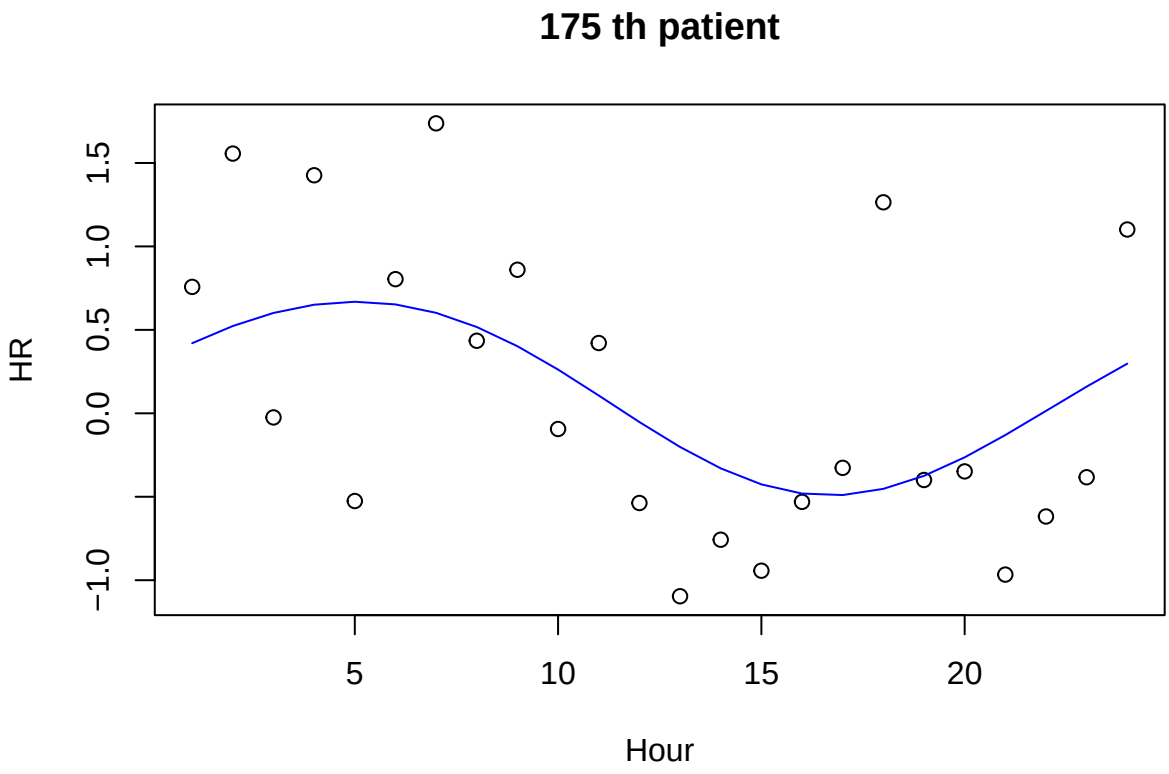



171 th patient
HR
2
1
0
−1
5
10
15
20
Hour


172 th patient
HR
0.5
0.0
−0.5
−1.0
5
10
15
20
Hour


173 th patient
HR
0.5
0.0
−0.5
−1.0
5
10
15
20
Hour


174 th patient
HR
0.0
−0.5
−1.0
−1.5
−2.0
5
10
15
20
Hour


175 th patient
HR
1.5
1.0
0.5
0.0
−1.0
5
10
15
20
Hour